

Online Resource ESM1: Wild bee species with total abundance and associated traits

Urban wild bees benefit from flower-rich anthropogenic land use depending on bee trait and scale

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Tab. ESM1.01: Wild bee species with total abundance and associated traits. Specimens without species identification were not grouped (-). Bumble bees were grouped in their own size class (B). *Hylaeus brevicornis* and *H. gredleri* could not be distinguished and were combined to *Hylaeus brevicornis* agg. *Andrena minutula* and *A. subopaca* were combined to *Andrena minutula* agg. Considering the controversial taxonomic status of some species (Schmidt et al. 2015), it was neither differentiated between *Nomada succincta* and *N. goodeniana* nor between *Andrena dorsata* and *A. propinqua*.

Species	Total Abundance	Size group	Foraging habit	Nesting habit
<i>Andrena bicolor</i>	3	medium	polylectic	below-ground
<i>Andrena carantonica</i>	10	large	polylectic	below-ground
<i>Andrena chrysosceles</i>	14	medium	polylectic	below-ground
<i>Andrena cineraria</i>	75	large	polylectic	below-ground
<i>Andrena clarkella</i>	2	large	oligolectic	below-ground
<i>Andrena dorsata</i>	10	medium	polylectic	below-ground
<i>Andrena flavipes</i>	106	medium	polylectic	below-ground
<i>Andrena fulva</i>	29	medium	polylectic	below-ground
<i>Andrena gravida</i>	15	large	polylectic	below-ground
<i>Andrena haemorrhoa</i>	287	medium	polylectic	below-ground
<i>Andrena helvola</i>	52	medium	polylectic	below-ground
<i>Andrena labialis</i>	1	large	oligolectic	below-ground
<i>Andrena minutula</i> agg.	18	small	polylectic	below-ground
<i>Andrena mitis</i>	2	medium	oligolectic	below-ground
<i>Andrena nigroaenea</i>	39	large	polylectic	below-ground
<i>Andrena nitida</i>	46	large	polylectic	below-ground
<i>Andrena ovatula</i>	1	medium	polylectic	below-ground
<i>Andrena praecox</i>	4	medium	oligolectic	below-ground
<i>Andrena strohmei</i>	8	small	polylectic	below-ground

<i>Andrena tibialis</i>	1	large	polylectic	below-ground
<i>Andrena vaga</i>	122	large	oligolectic	below-ground
<i>Andrena varians</i>	1	medium	polylectic	below-ground
<i>Andrena ventralis</i>	7	small	oligolectic	below-ground
<i>Andrena sp.</i>	2	-	-	-
<i>Anthidium manicatum</i>	2	large	polylectic	both
<i>Bombus hortorum</i>	3	B	polylectic	both
<i>Bombus hypnorum</i>	7	B	polylectic	above-ground
<i>Bombus lapidarius</i>	9	B	polylectic	both
<i>Bombus lucorum</i>	17	B	polylectic	below-ground
<i>Bombus pascuorum</i>	44	B	polylectic	both
<i>Bombus pratorum</i>	7	B	polylectic	both
<i>Bombus soroensis</i>	2	B	polylectic	below-ground
<i>Bombus terrestris</i>	84	B	polylectic	both
<i>Bombus vestalis</i>	7	B	parasitic	parasitic
<i>Chelostoma florissomne</i>	3	medium	oligolectic	above-ground
<i>Chelostoma rapunculi</i>	3	medium	oligolectic	above-ground
<i>Colletes cunicularius</i>	7	large	polylectic	below-ground
<i>Colletes daviesanus</i>	2	medium	oligolectic	above-ground
<i>Colletes fodiens</i>	3	medium	oligolectic	below-ground
<i>Colletes similis</i>	1	small	oligolectic	below-ground
<i>Dasypoda hirtipes</i>	41	large	oligolectic	below-ground
<i>Halictus quadricinctus</i>	4	large	polylectic	below-ground
<i>Halictus rubicundus</i>	3	medium	polylectic	below-ground
<i>Halictus scabiosae</i>	22	large	polylectic	below-ground
<i>Halictus subauratus</i>	3	small	polylectic	below-ground
<i>Halictus tumulorum</i>	2	small	polylectic	below-ground
<i>Hoplitis adunca</i>	1	large	oligolectic	above-ground
<i>Hoplitis leucomelana</i>	1	small	polylectic	above-ground
<i>Hylaeus angustatus</i>	1	small	polylectic	above-ground
<i>Hylaeus brevicornis aggr.</i>	5	small	polylectic	above-ground
<i>Hylaeus communis</i>	28	small	polylectic	above-ground
<i>Hylaeus confusus</i>	5	small	polylectic	above-ground
<i>Hylaeus cornutus</i>	1	small	polylectic	above-ground
<i>Hylaeus dilatatus</i>	5	small	polylectic	above-ground
<i>Hylaeus hyalinatus</i>	7	small	polylectic	above-ground
<i>Hylaeus nigrinus</i>	13	small	oligolectic	above-ground
<i>Hylaeus paulus</i>	1	small	polylectic	above-ground
<i>Hylaeus sp.</i>	1	-	-	-
<i>Lasioglossum calceatum</i>	6	medium	polylectic	below-ground
<i>Lasioglossum fulvicorne</i>	26	small	polylectic	below-ground
<i>Lasioglossum intermedium</i>	5	small	polylectic	below-ground
<i>Lasioglossum laticeps</i>	2	small	polylectic	below-ground
<i>Lasioglossum leucopus</i>	1	small	polylectic	below-ground
<i>Lasioglossum leucozonium</i>	6	medium	polylectic	below-ground
<i>Lasioglossum malachurum</i>	2	medium	polylectic	below-ground

<i>Lasioglossum minutissimum</i>	7	small	polylectic	below-ground
<i>Lasioglossum monstificum</i>	5	small	polylectic	below-ground
<i>Lasioglossum morio</i>	22	small	polylectic	below-ground
<i>Lasioglossum nitidiusculum</i>	3	small	polylectic	below-ground
<i>Lasioglossum nitidulum</i>	3	small	polylectic	below-ground
<i>Lasioglossum pauxillum</i>	76	small	polylectic	below-ground
<i>Lasioglossum quadrinotatum</i>	2	small	polylectic	below-ground
<i>Lasioglossum rufitarse</i>	1	small	polylectic	below-ground
<i>Lasioglossum semilucens</i>	1	small	polylectic	below-ground
<i>Lasioglossum sexnotatum</i>	3	medium	polylectic	below-ground
<i>Lasioglossum sexstrigatum</i>	5	small	polylectic	below-ground
<i>Lasioglossum villosulum</i>	6	small	polylectic	below-ground
<i>Lasioglossum sp.</i>	2	-	-	-
<i>Macropis fulvipes</i>	1	medium	oligolectic	below-ground
<i>Megachile ericetorum</i>	2	large	oligolectic	above-ground
<i>Megachile ligniseca</i>	1	large	polylectic	above-ground
<i>Megachile maritima aff.</i>	2	large	-	-
<i>Megachile versicolor</i>	3	large	polylectic	above-ground
<i>Megachile willughbiella</i>	1	large	polylectic	both
<i>Nomada alboguttata</i>	2	medium	parasitic	parasitic
<i>Nomada fabriciana</i>	10	medium	parasitic	parasitic
<i>Nomada ferruginata</i>	2	medium	parasitic	parasitic
<i>Nomada flava</i>	1	medium	parasitic	parasitic
<i>Nomada flavoguttata</i>	1	small	parasitic	parasitic
<i>Nomada fucata</i>	4	medium	parasitic	parasitic
<i>Nomada lathburiana</i>	7	large	parasitic	parasitic
<i>Nomada panzeri</i>	13	medium	parasitic	parasitic
<i>Nomada ruficornis</i>	6	medium	parasitic	parasitic
<i>Nomada succincta</i>	1	large	parasitic	parasitic
<i>Osmia bicolor</i>	17	large	polylectic	above-ground
<i>Osmia bicornis</i>	117	medium	polylectic	above-ground
<i>Osmia brevicornis</i>	1	medium	oligolectic	above-ground
<i>Osmia cornuta</i>	2	large	polylectic	above-ground
<i>Osmia spinulosa</i>	2	small	oligolectic	above-ground
<i>Panurgus banksianus</i>	1	large	oligolectic	below-ground
<i>Panurgus calcaratus</i>	1	small	oligolectic	below-ground
<i>Sphecodes albilabris</i>	1	large	parasitic	parasitic
<i>Sphecodes crassus</i>	3	small	parasitic	parasitic
<i>Sphecodes ephippius</i>	1	small	parasitic	parasitic
<i>Sphecodes puncticeps</i>	2	small	parasitic	parasitic

References

Schmidt S, Schmid-Egger C, Morinière J, Haszprunar G, Hebert PDN (2015) DNA barcoding largely supports 250 years of classical taxonomy: identifications for Central European bees (Hymenoptera, Apoidea partim). *Mol Ecol Resour*, 15(4):985–1000.
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